

A disproof of Conjecture 00000004007

The p -variation is 1-homogeneous, so $\|X + X\| = 2\|X\| > 2^{1/q}\|X\|$

1 The conjecture

Conjecture 00000004007 reads (English):

Definition: The p -variation is the p -th root of the supremum over partitions of the sum of p -th powers of the increments. *Conjecture:* The p -variation satisfies the optimal subadditivity inequality $\|X+Y\| \leq (\|X\|^q + \|Y\|^q)^{1/q}$, where q is the conjugate exponent of p , and the inequality cannot be enlarged.

The conjecture is a conjunction (the inequality holds, and it is optimal). We show that the inequality itself fails, for every $p \in (1, \infty)$ and also for $p = 1$.

2 Definitions and conventions

Definition 1 (Partition). A partition of $[a, b]$ is a finite sequence $a = t_0 < t_1 < \dots < t_n = b$ (Lean: `Partition a b`, fields `n, t : Fin (n+1) -> R`, `StrictMono t`, `t 0 = a`, `t (last n) = b`).

Definition 2 (p -variation). For a path $X : \mathbb{R} \rightarrow E$ (E a real normed space; only the values on $[a, b]$ matter) and $p > 0$,

$$V_p(X; [a, b]) = \left(\sup_{\pi} \sum_{i=1}^n \|X(t_i) - X(t_{i-1})\|^p \right)^{1/p} \in [0, \infty],$$

the supremum over all partitions π of $[a, b]$ (Lean: `incrSum` and `pVar`).

Conventions. Values are taken in $[0, \infty]$ (Lean `ENNReal`), so the supremum always exists and $\infty^{1/p} = \infty$; our counterexamples have finite values, so this choice is not used. The conjugate exponent of $p \in (1, \infty)$ is the $q > 0$ with $1/p + 1/q = 1$, i.e. $q = p/(p-1) \in (1, \infty)$ (Mathlib's `Real.HolderConjugate p q`, which requires $p, q > 0$ and $p^{-1} + q^{-1} = 1$). For $p = 1$ we take $q = \infty$ and read $(a^q + b^q)^{1/q}$ as $\max(a, b)$. Paths are real-valued in the counterexample; a real-valued counterexample is also one for paths in $\mathbb{R}^d$ (embed in the first coordinate), and Lemma 4 holds in every real normed space.

3 Proof

Lemma 3 (Homogeneity; `pVar_smul`). For $p > 0$, $c \in \mathbb{R}$ and every path X : $V_p(cX; [a, b]) = |c| V_p(X; [a, b])$.

Proof. For each partition the increments of cX are c times those of X , so each sum is multiplied by $|c|^p$ (`incrSum_smul`). Since $\sup_{\pi} |c|^p S_{\pi} = |c|^p \sup_{\pi} S_{\pi}$ in $[0, \infty]$, and $(|c|^p s)^{1/p} = |c| s^{1/p}$, the claim follows. $\square$

Lemma 4 (Failure on the diagonal; `fails_at_diagonal`, `fails_at_diagonal_one`). *Let $0 < V_p(X; [a, b]) =: v < \infty$. If $1/p + 1/q = 1$ with $p, q > 0$, then $(v^q + v^q)^{1/q} = 2^{1/q} v < 2v = V_p(X + X; [a, b])$. If $p = 1$, then $\max(v, v) = v < 2v = V_1(X + X; [a, b])$.*

Proof. $X + X = 2X$, so the right-hand value is $2v$ by homogeneity. Since $q > 1$, $1/q < 1$ and $2^{1/q} < 2$; multiply by $0 < v < \infty$. $\square$

Lemma 5 (A non-degenerate path; `iSup_incrSum_id`, `pVar_id`). *Let $p \geq 1$ and $b - a = 1$. For $X(t) = t$, the supremum of the increment sums over partitions of $[a, b]$ is 1; in particular $V_p(X; [0, 1]) = 1$.*

Proof. Each increment $\Delta_i = t_i - t_{i-1}$ lies in $[0, 1]$, so $\Delta_i^p \leq \Delta_i$ (as $p \geq 1$) and $\sum \Delta_i^p \leq \sum \Delta_i = b - a = 1$ (telescoping, `sum_succ_sub`). The two-point partition $a < b$ gives exactly 1. $\square$

Theorem 6 (`conjecture_00000004007_false`, `conjecture_00000004007_false'`). *For every $p \in (1, \infty)$ with conjugate exponent q , the continuous paths $X = Y = (t \mapsto t)$ on $[0, 1]$ satisfy*

$$V_p(X + Y; [0, 1]) = 2 > 2^{1/q} = (V_p(X; [0, 1])^q + V_p(Y; [0, 1])^q)^{1/q}.$$

Theorem 7 ($p = 1$; `conjecture_00000004007_false_one`). *For $p = 1$, $q = \infty$: $V_1(X + Y; [0, 1]) = 2 > 1 = \max(V_1(X), V_1(Y))$ for the same paths.*

Remark 8 (Other readings refuted in Lean). (i) *Raw supremum* (`raw_version_false`): if $\|X\|$ denotes the supremum without the p -th root, it is p -homogeneous, the left side at $Y = X$ is $2^p v$ and the right side is $2^{1/q} v$, and $1/q < 1 < p$. (ii) *Concatenation* (`concat_version_false`): if $X + Y$ denotes the concatenation of X on $[0, 1]$ and Y on $[1, 2]$ (agreeing at $t = 1$), take $Z(t) = t$ on $[0, 2]$: both pieces have p -variation 1 while $V_p(Z; [0, 2]) \geq 2$ by the two-point partition, and $2 > 2^{1/q}$.

4 Scope

Refuted: the inequality for every $p \in (1, \infty)$ with $q = p/(p - 1)$; for $p = 1$ with $q = \infty$ read as the maximum; with or without the p -th root; with $X + Y$ the pointwise sum or the concatenation. Since the inequality is false, the clause “cannot be enlarged” is moot. Not addressed: $0 < p < 1$ (the conjugate exponent would be negative, and continuous paths of finite p -variation are then constant) and $p = \infty$ (no p -th root; $q = 1$ and the inequality would reduce to the triangle inequality for the oscillation, which holds). The counterexample is not degenerate: $X(t) = t$ is continuous, non-constant and has p -variation exactly 1.

5 Relation to the earlier submission

The same mathematical argument ($X = Y$, homogeneity, $2 > 2^{1/q}$) was merged as PR #42 (earthking11) and removed in the re-audit of commit 541cf4fb with the reason “pure powers-of-2 arithmetic; p -variation unformalized”. Its Lean file stated only facts such as $\neg(2^2 \leq 2)$ and $2 < 2^q$ over the natural numbers, and its docstring said “There is no real-valued p -variation formalised here”. The present Lean file defines partitions and the p -variation as the p -th root of the supremum

of the increment sums, proves homogeneity and the diagonal failure for every path of finite nonzero p -variation in every real normed space, computes the p -variation of $t \mapsto t$ exactly, and states the conjecture's inequality with Mathlib's conjugate-exponent predicate for every real $p > 1$ (not only integer instances).

6 Lean formalization

File `lean/Conjecture4007/Basic.lean` (Lean 4.33.1, Mathlib), namespace `Tlmc4007`. Main theorems: `conjecture_00000004007_false` (for all p, q with `p.HolderConjugate q`), `conjecture_00000004007_false'` (for all $p > 1$ with $q = p/(p - 1)$), `conjecture_00000004007_false_one`, `raw_version_false`, `concat_version_false`; supporting `pVar_smul`, `fails_at_diagonal`, `pVar_id`. `lake build` succeeds; `#print axioms` of each main theorem gives `[propext, Classical.choice, Quot.sound]`. No sorry, no `native_decide`, no added axioms.